On Sampling Error in Batch Action-Value Prediction Algorithms

Brahma S. Pavse
Salesforce
UT Austin

Josiah P. Hanna
School of Informatics
The University of Edinburgh

Ishan Durugkar
UT Austin

Peter Stone
UT Austin
Sony AI

Learning Agents Research Group
The University of Texas at Austin

Abstract

- In offline RL, **sampling error** arises due to **variance in sampling** from a finite batch of data, resulting in a **distribution mismatch** between the **empirical distribution** of the data and the **sampling distribution** of the data.
- We study this problem by analyzing the **on-policy offline** RL setting for **action-value function estimation**.
- We give convergence proofs for the considered action-value learning algorithms and relate the algorithms to their **certainty-equivalence estimates.**

Algorithms

1. Batch Action-Value (AV)-TD

$$q(s,a) \leftarrow q(s,a) + [R + \boxed{\gamma q(s',a')} - q(s,a)]$$

2. Batch Expected AV-TD

$$q(s,a) \leftarrow q(s,a) + [R + \boxed{\gamma \sum_{a'} \pi(a'|s) q(s',a')} - q(s,a)]$$

3. Batch PSEC AV-TD

$$q(s,a) \leftarrow q(s,a) + \boxed{\frac{\pi(a'|s')}{\hat{\pi}(a'|s,a,s')}} [R + \gamma q(s',a') - q(s,a)]$$

Accounting for ***sampling error*** *in offline RL leads to* ***more accurate*** *action-value learning.*

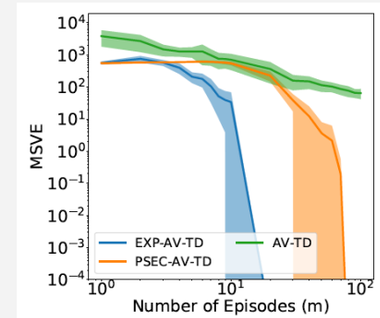
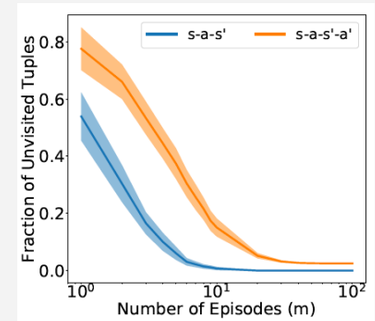
Paper Link

Theoretical Results

- Asymptotically, batch expected AV-TD and batch PSEC AV-TD **correct for all policy sampling error.**
- Asymptotically, batch expected AV-TD and batch PSEC AV-TD converge to the **same** fixed point.
- Convergence of all the considered algorithms is related to the so-called **certainty-equivalence estimate.**

Empirical Results (Tabular Deterministic Gridworld)

Batch EXP-AV-TD and PSEC-AV-TD **outperform** the fundamental (AV-TD) algorithm for **all batch sizes.**



Batch EXP-AV-TD outperforms PSEC-AV-TD for smaller batch sizes since the former corrects for all policy sampling error when all **state-action-next-states** are visited, while the latter can do the same when all **state-action-next-state-next-actions** are visited.